Statistical Theory

The Relationship Between Movie Genre and Inflation-Adjusted Box Office Revenue for Disney Films

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Abstract - Movie genre is often considered one of the factors associated with a film's commercial success. This study examines whether inflation-adjusted box office revenue differs across Disney movie genres using a dataset of 579 films released between 1937 and 2016. After excluding movies with missing genre information, 562 films were included in the primary analysis. Since the revenue distributions were highly right-skewed and variances differed across genres, Welch's ANOVA and the non-parametric Kruskal-Wallis test were used, followed by Games-Howell post-hoc comparisons. The analysis revealed statistically significant differences in inflation-adjusted revenue across genres. Adventure films consistently outperformed several lower-revenue genres, while Musical films showed the highest mean revenue due to a small number of exceptionally successful titles. As a secondary analysis, a genre-based classification model showed only modest predictive performance, indicating that genre alone is not sufficient to reliably predict blockbuster success. Overall, the findings suggest that movie genre is significantly associated with box office revenue, although it is only one of several factors that influence a film's commercial success.

The code for this project is available at:

Introduction

This study examines the association between movie genre and commercial success using a dataset of Disney films released between 1937 and 2016. To ensure economic comparability across historical periods, inflation-adjusted box office revenue serves as the continuous outcome variable, analyzed against genre as the primary categorical predictor. Following data preprocessing-which included retaining four valid zero-revenue observations but excluding 17 films lacking genre classification-the final analytical sample comprises 562 movies.

The main research question is:

Are inflation-adjusted box office revenues significantly different across Disney movie genres?

The hypotheses are:

H_0: There is no statistically significant difference in inflation-adjusted box office revenue across Disney movie genres.

H_1: At least one Disney movie genre differs significantly in inflation-adjusted box office revenue.

Results

The analysis began with an overview of the dataset and the distribution of the variables. A total of 579 Disney films were included in the original dataset, of which 562 had complete genre information and were used for the primary statistical analysis. Table 1 summarizes the overall characteristics of the dataset.

Statistic	Value
Number of films	579
Known genre	562
Missing genre	17 (2.9%)
Mean revenue	$118.76 million
Median revenue	$55.16 million
Standard deviation	$286.09 million
First quartile	$22.74 million
Third quartile	$119.20 million
Maximum	$5,228.95 million

Table 1. Summary statistics of the Disney movie dataset.

The mean revenue is more than twice the median revenue, indicating a strongly right-skewed distribution. This suggests that a relatively small number of highly successful movies substantially increase the average revenue, while most films earned considerably less.

Genre Distribution

Before comparing revenues across genres, we first examined the distribution of movies among the different genre categories.

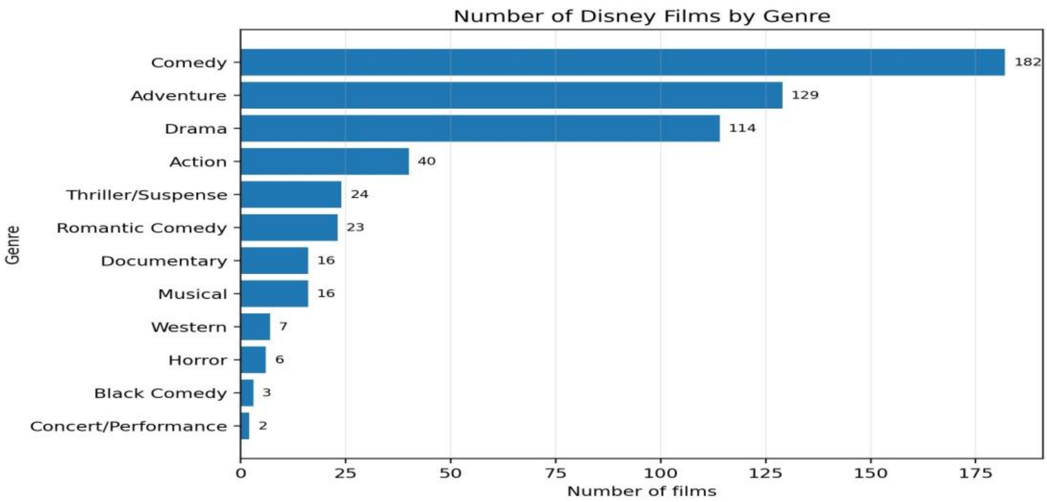

Figure 1. Genre sample sizes among the 562 films with known genre.

As shown in Figure 1, Comedy and Adventure contain the largest number of movies, whereas genres such as Black Comedy, Concert/Performance, Horror and Western contain relatively few observations. These differences in sample size were taken into account when selecting the statistical methods used in the analysis.

Genre-Level Descriptive Statistics

To better understand how box office revenue varies across genres, descriptive statistics were calculated for each genre. Since the overall revenue distribution is highly skewed, both the mean and the median are presented.

Genre	n	Mean	Median	IQR	SD
Musical	16	603.60	103.17	204.40	1,346.57
Adventure	129	190.40	102.25	208.15	254.60
Western	7	73.82	89.04	35.99	36.38
Action	40	137.47	69.01	144.65	145.31
Romantic Comedy	23	77.78	57.93	68.74	79.86
Concert/ Performance	2	57.41	57.41	19.24	27.20
Thriller/Suspense	24	89.65	51.80	83.36	112.12
Black Comedy	3	52.24	51.58	24.18	24.19
Comedy	182	84.67	51.20	84.14	122.65
Drama	114	71.89	39.33	73.33	146.11
Horror	6	23.41	18.59	11.36	13.93
Documentary	16	12.72	12.34	17.67	11.34

Table 2. Revenue by genre in USD millions, ordered by median revenue.

Table 2 shows substantial differences in inflation-adjusted revenue across movie genres. Musical films have the highest average revenue ($603.60 million), whereas Documentary and Horror films have the lowest. However, the large gap between the mean and median for Musicals indicates this high average is driven by a few extreme outliers. In contrast, Adventure films achieve more consistent commercial performance, sharing a similar median with Musicals but presenting a considerably lower mean. While these descriptive statistics suggest revenue differs across genres, formal statistical tests were performed to evaluate the significance of these observed differences.

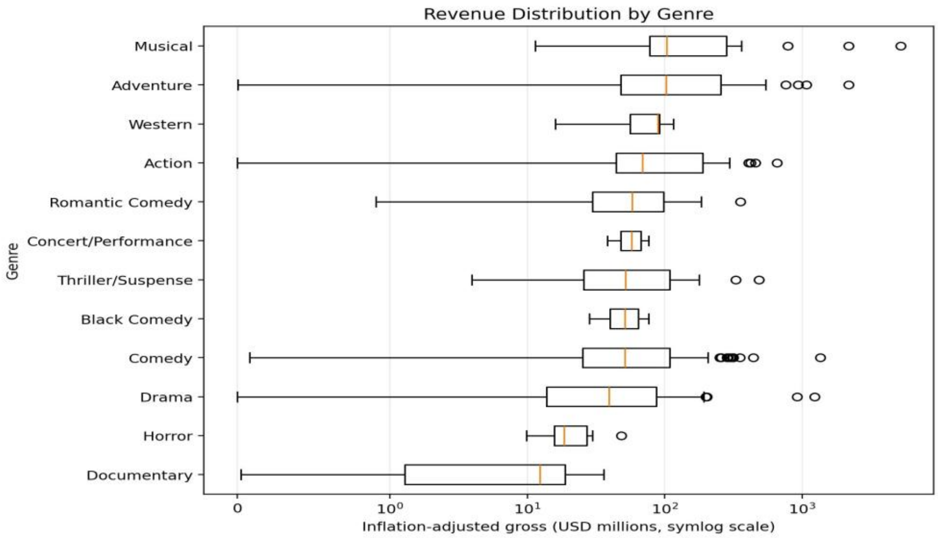

Figure 2. Boxplots of inflation-adjusted revenue by genre. A symmetric-log scale preserves zero values while making the extreme range readable.

Figure 2 illustrates the revenue distribution within each genre. Most genres are right-skewed and contain extreme outliers. Musical films show particularly high variability, whereas Adventure films exhibit a more consistent distribution.

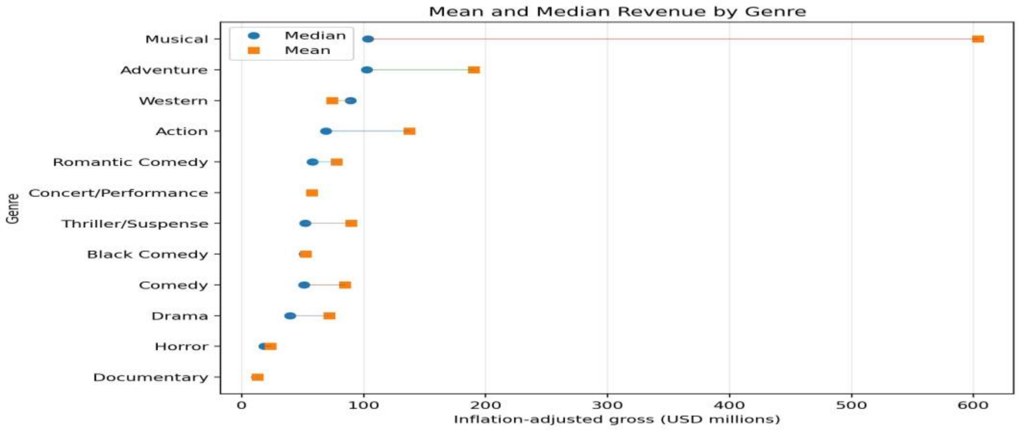

Figure 3. Comparison of genre means and medians. Large separations signal strong skew and influential outliers.

Figure 3 compares the mean and median revenue for each genre. The large gap observed for Musicals reflects the influence of a few extreme values, whereas Adventure shows more consistent revenue.

Before comparing revenue across genres, we evaluated whether the assumptions required for classical one-way ANOVA were satisfied. The Shapiro-Wilk test indicated that seven of the eleven eligible genres deviated significantly from normality at the 5% significance level. The Concert/Performance genre was excluded from this assessment because it contains only two films. In addition, the Brown-Forsythe test indicated unequal variances across genres ($F = 5.48$, $p = 2.46 \times 10^{-8}$). Together, these findings suggest that the assumptions of the classical ANOVA are not fully satisfied. Therefore, Welch's ANOVA was selected as the primary parametric test, while the Kruskal-Wallis test was used as a complementary non-parametric analysis.

Assumption diagnostics and omnibus tests

Test	Statistic	p-value	Interpretation
Classical one-way ANOVA	F(11, 550) = 6.23	1.02 x 10^-9	Reject H0
Welch's ANOVA	F(11, 24.39) = 13.53	8.02 x 10^-8	Reject H0
Kruskal-Wallis	H(11) = 88.92	2.71 x 10^-14	Reject H0
Brown-Forsythe variance test	F(11, 550) = 5.48	2.46 x 10^-8	Unequal variances

Table 3. Omnibus tests for genre differences and variance equality.

Both parametric (Welch's ANOVA) and non-parametric (Kruskal-Wallis) tests consistently indicate that inflation-adjusted revenue differs significantly across genres. Effect sizes ($\eta^2 = 0.111$ and $\varepsilon^2 = 0.142$) show a moderate association, though substantial variation remains unexplained by genre alone.

Post-hoc Comparisons Games-Howell post-hoc tests, accounting for unequal variances and sample sizes, identified 13 significant pairwise differences. Adventure yielded significantly higher revenues than seven other genres (Documentary, Horror, Drama, Black Comedy, Comedy, Western, and Romantic Comedy). Action also outperformed Documentary and Horror, while Comedy and Drama surpassed Documentary. Notably, despite having the highest average revenue, Musicals showed no significant differences. This is likely due to the genre's small sample size (n=16) and high variability, which severely reduced statistical power.

Higher-mean genre	Comparison genre	Mean difference	Adjusted p	95% CI
Adventure	Documentary	177.68	<0.001	[102.49, 252.87]
Adventure	Horror	166.98	<0.001	[90.02, 243.95]
Adventure	Drama	118.50	0.0006	[31.69, 205.32]
Adventure	Black Comedy	138.15	0.0012	[42.60, 233.71]
Adventure	Comedy	105.73	0.0013	[25.56, 185.90]
Adventure	Western	116.58	0.0022	[27.19, 205.98]
Adventure	Romantic Comedy	112.62	0.0055	[19.39, 205.85]

Table 4. Significant Games-Howell comparisons involving Adventure (USD millions). Full significant results appear in Appendix A.

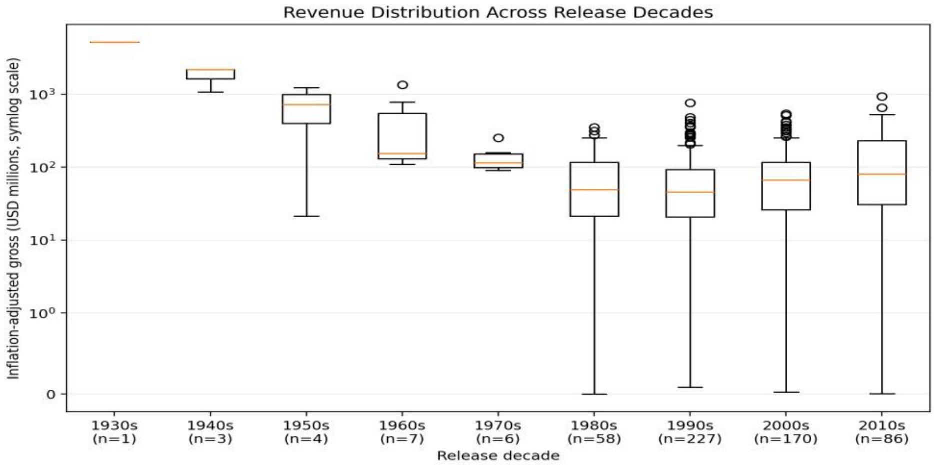

Figure 4. Revenue distributions by release decade. Historical period is a plausible confounder that genre only comparisons do not control.

Figure 4 presents the distribution of inflation-adjusted revenue across release decades. Although inflation adjustment improves comparability over time, noticeable differences remain between historical periods. Earlier decades contain relatively few films, whereas the 1990s, 2000s, and 2010s include substantially larger samples. This suggests that release period may influence box office performance and should be considered as a potential confounding factor when interpreting genre differences.

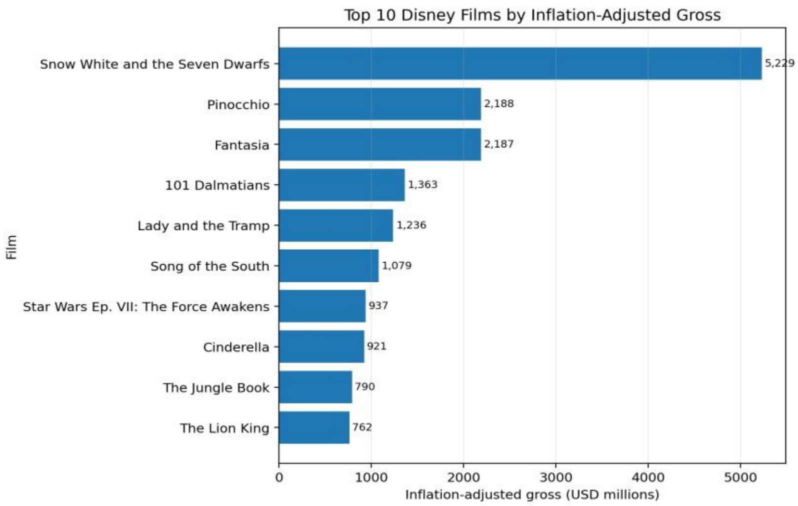

Figure 5. The ten highest inflation-adjusted grossing films. A few historical successes strongly influence genre means.

Figure 5 illustrates that the ten highest-grossing Disney films heavily skew the average revenues of their respective genres, particularly Musicals, which aligns with the previously noted mean-median gap.

To further evaluate genre's predictive power, two additional analyses were conducted. A robust OLS regression on log-transformed revenue explained only ~11% of the variance ($R^2 = 0.108$). Similarly, a logistic regression classifying top-quartile films based solely on genre achieved a modest cross-validated ROC AUC of 0.666. Collectively, these models indicate that while genre is significantly associated with commercial success, it is insufficient on its own to reliably predict a blockbuster, as most variance is driven by unmeasured factors.

Metric	Value
Top-quartile prevalence	25.1%
Cross-validated ROC AUC	0.666
Balanced-model accuracy	68.5%
Balanced accuracy	64.6%
Precision	40.8%
Recall	56.7%
F1 score	47.5%

Table 5. Five-fold cross-validated performance of the genre-only blockbuster classifier.

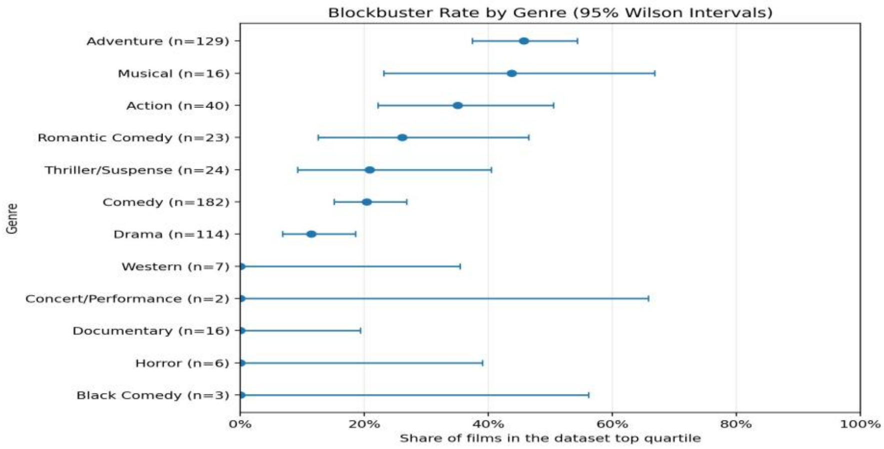

Figure 6. Proportion of films in the top revenue quartile by genre. Wide intervals for small genres show substantial uncertainty.

Figure 6 shows the proportion of top-quartile films by genre. Wider confidence intervals for smaller genres indicate greater uncertainty and should be interpreted with caution.

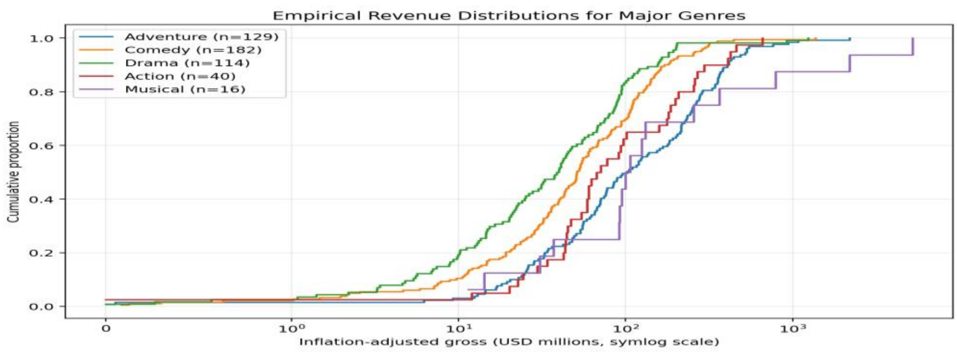

Figure 7. Empirical cumulative distributions for the five focal genres. Curves farther to the right indicate generally larger revenues.

Figure 7 compares the ECDFs of the five main genres. Adventure generally achieves higher revenues more consistently, whereas the Musical distribution is more strongly influenced by a few extreme values.

Methods

All analyses were performed in Python using standard scientific libraries for statistical analysis and data visualization. After preprocessing the dataset and removing observations with missing genre information, descriptive statistics and visualizations were generated. Normality was assessed using the Shapiro-Wilk test, and homogeneity of variances was evaluated using the Brown-Forsythe test. Because the assumptions of classical ANOVA were violated, Welch's ANOVA and the Kruskal-Wallis test were used, followed by Games-Howell post-hoc comparisons. In addition, robust linear regression and logistic regression were performed to evaluate the predictive value of movie genre.

Discussion

The analysis confirms a robust, statistically significant relationship between Disney movie genre and inflation-adjusted revenue, supported consistently across classical ANOVA, Welch's ANOVA, and Kruskal-Wallis tests. Crucially, while Musicals exhibit the highest mean revenue, this is heavily skewed by historic outliers (e.g., *Snow White*, *Pinocchio*, and *Fantasia*). A more accurate interpretation reveals that Adventure and Musical films share nearly identical medians, though Adventure remains the most consistently high-performing genre in post-hoc testing. Finally, while genre is statistically informative, regression and classification models demonstrate its limited predictive power for blockbuster success. A comprehensive predictive model would require additional variables, such as production budgets, marketing expenditures, franchise status, and release timing. Therefore, statistical significance should not be interpreted as strong predictive ability.

Limitations

- The dataset includes Disney films only and ends in 2016, so the results should not be generalized to the entire film industry or to the current streaming era.
- Genre sample sizes are highly unequal. Comedy has 182 films, whereas Concert/Performance has only two and Black Comedy has three.
- Inflation adjustment addresses currency value but not changes in population, ticket prices, theatrical windows, re-releases, distribution, or international coverage.
- The analysis does not include production budgets or marketing costs; therefore, gross revenue is not the same as profitability.
- Seventeen films lack genre labels. Excluding them avoids treating missingness as a real genre but may introduce minor selection effects.
- Observational comparisons show association, not causation. Genre may be correlated with release period, budget, franchise status, and other omitted variables.

Conclusion

Disney films show statistically significant revenue differences across genres. Adventure films perform strongly and consistently, while the unusually large Musical mean is driven by a small number of extreme successes. Genre has a moderate association with revenue but is not sufficient by itself to explain or predict blockbuster performance. Future analyses should combine genre with production budget, marketing, franchise status, ratings, release timing, and historical period.

References

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4. Raw data mirror: https://raw.githubusercontent.com/maliat-hossain/FileProcessing/main/disney_movies_total_gross.csv
5. Welch, B. L. (1951). On the comparison of several mean values: an alternative approach. Biometrika, 38(3/4), 330-336.

Appendix A. Full Significant Games-Howell Results

Higher-mean genre	Lower-mean genre	Difference	Adjusted p	95% CI
Adventure	Documentary	177.68	<0.001	[102.49, 252.87]
Comedy	Documentary	71.95	<0.001	[40.45, 103.45]
Adventure	Horror	166.98	<0.001	[90.02, 243.95]
Comedy	Horror	61.25	<0.001	[24.62, 97.89]
Action	Documentary	124.76	0.0002	[44.50, 205.01]
Adventure	Drama	118.50	0.0006	[31.69, 205.32]
Action	Horror	114.06	0.0010	[32.29, 195.83]
Adventure	Black Comedy	138.15	0.0012	[42.60, 233.71]
Adventure	Comedy	105.73	0.0013	[25.56, 185.90]
Adventure	Western	116.58	0.0022	[27.19, 205.98]
Drama	Documentary	59.18	0.0025	[12.60, 105.75]
Adventure	Romantic Comedy	112.62	0.0055	[19.39, 205.85]
Romantic Comedy	Documentary	65.06	0.0300	[3.97, 126.15]

Table A1. All Games-Howell comparisons significant at alpha = 0.05 (USD millions).

Appendix B. Sensitivity Analysis and Corrections to the Original Draft

The original draft included the 17 missing genre values as an "Unknown" group. Reproducing that exact choice yields classical ANOVA F = 6.06, p = 4.85 x 10^-10 and Kruskal-Wallis H = 108.01, p = 1.48 x 10^-17. The revised primary analysis excludes "Unknown" because it is not a true genre, and the conclusion remains unchanged.

Excluding the four zero-revenue observations also leaves the result unchanged: Welch F = 13.60, p = 7.65 x 10^-8 and Kruskal-Wallis H = 89.82, p = 1.81 x 10^-14.

Major revisions made in this version include: restoring the missing figures; correcting the Thriller/Suspense label; distinguishing mean from median; replacing a sole reliance on standard ANOVA with Welch's ANOVA and Kruskal-Wallis; documenting unequal variances; reporting effect sizes and post-hoc results; evaluating the regression claims quantitatively; clarifying missing-data handling; and expanding the limitations and visual analysis.